SkyNet TSO Filings Module — Implementation Plan v0.1

Oct 29, 2026 · @Matthew Bowers

Purpose and scope

The TSO filings module gives SkyNet a system of record for Skybolt's FAA certifications: which drawings are approved under which authorization, at which revision, submitted when, accepted when, by whom. Today that record is Ned's binder, a folder tree on the common drive with a dozen copies of everything, and Ned's memory. The binder is good; it is also a single copy held by one person, and it has already produced two Revision O's, a missing F through I, and a V that became V2.

What the module does:

- Holds every certification Skybolt owns (TSO-C148 authorization, seven STCs, two PMA numbers) and the controlled drawings that belong to each.
- Tracks every revision of every drawing, and every Master Drawing List revision as a dated snapshot of which drawing revisions were current.
- Records every submission to the FAA and every FAA response, with the documents attached, so the file of record is complete and searchable.
- Prevents the housekeeping errors a paper system invites: duplicate revision letters, a drawing revised in SkyNet but never added to a submission, a submission with no FAA response after six months.
- Generates the recurring paperwork from the data: the submission letter, the Certificate of Conformance, the MDL itself, and the customer approval package, so they are always consistent with each other.
- Links each controlled drawing to the SkyNet part (and through it to the Fishbowl part number) so a part's certification status is visible wherever the part appears.

What it is not: it is not a drawing management or CAD system (drawings remain the engineer's files; the module holds the released PDF and the revision record), it does not replace Ned's or Roger's judgment on what is a minor versus major change, and nothing it generates goes to the FAA without a person signing it. Scope for v1 is the TSO-C148 authorization; the STCs and PMA are modeled from the start so they can be loaded in the next phase without schema change.

It lives in SkyNet (Supabase/PostgreSQL, React) as a Compliance-owned tab, built on TEST first and promoted to PROD by a person after sign-off, per the standing rule.

What the binder teaches about the data

Ned's binder is already a database; it is just on paper. Reading it end to end, five record types account for everything in it, and the way they relate is the schema.

Record	What it is in the binder	Count today	Key facts it carries
Certification	The TSOA itself; the STCs and PMA numbers in the STC archive	1 TSOA, 7 STCs, 2 PMA numbers	Type, number, standard, issuing office, dates, limitations, status
Controlled drawing	One binder section per MDL page: a drawing with its own revision block	87 on MDL Rev V	Drawing number, title, product family, MDL page number, part it defines
Drawing revision	One row of a drawing's revision block: letter, description, by, date	Roughly 500 rows across 87 drawings	Letter, date, description, author (NCB), whether it has been submitted and accepted
MDL revision	One Master Drawing List issue: a dated snapshot listing each drawing at a revision, with notes	D through V, 19 issues (one letter used twice)	Letter, date, nature (original, authorization, minor), list of drawing revisions, asterisks for changed items
FAA event	A submission letter going out, or an FAA letter, email, or stamp coming back	About 20 submissions, about 20 responses, 2002 to 2026	Direction, date, FAA project number, reviewer, office, what it covers, outcome, attached documents

Three relationships carry the meaning:

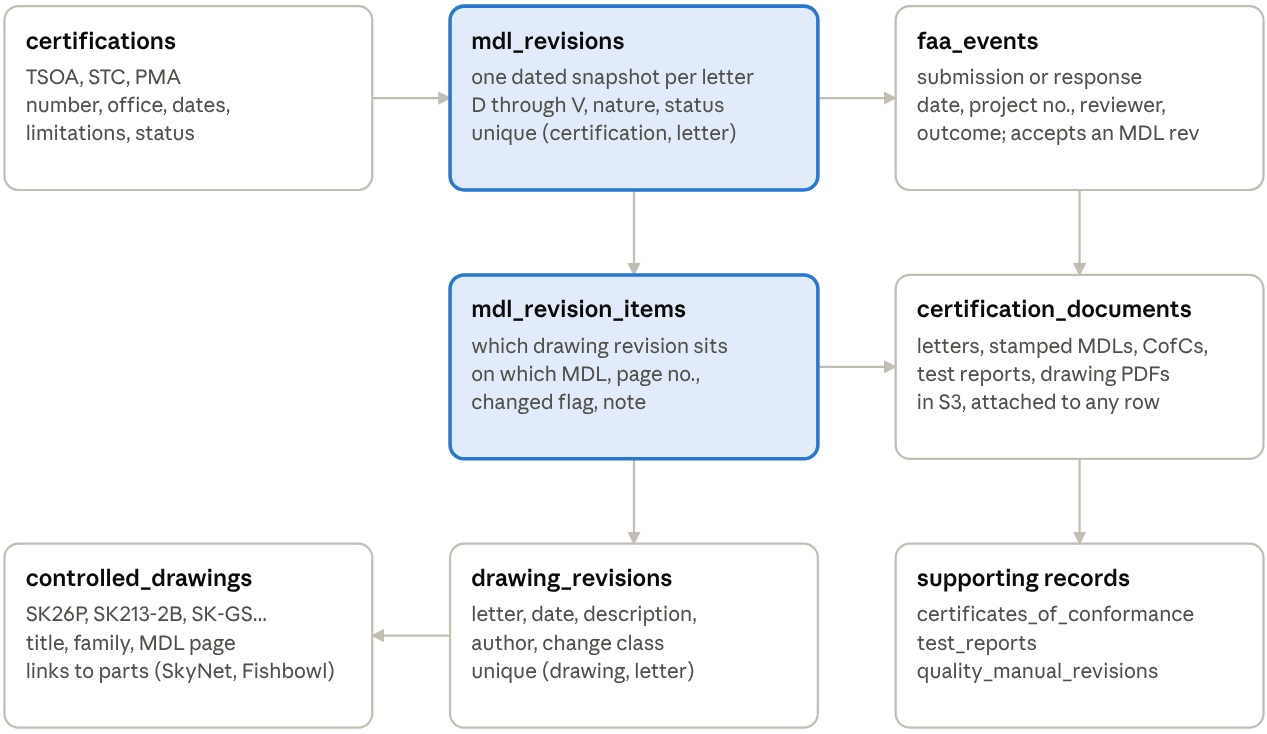
- A drawing revision is approved when it appears on an MDL revision that the FAA has accepted. Approval is a property of the pair (drawing revision, accepted MDL revision), not of the drawing alone. This is why the MDL is the controlling document and why the database must store MDL membership explicitly rather than deriving "current revision" from the drawing table.
- An MDL revision is submitted by one or more FAA events (a letter, sometimes followed by supplementary letters as with Rev L in 2009) and accepted by one FAA event (a stamp, a letter, or an email). Until the accepting event exists, the MDL revision and every drawing revision new to it are pending.
- Supporting records hang off the MDL revision: the Certificate of Conformance reissued with it, the test report that substantiated a new design, the quality manual revision in force at the time.

The binder also shows what the paperwork looks like when it is right. From Revision O on, the submission is a one-page letter with a fixed sentence citing 21.619(a), a CofC dated the same day, the MDL with changed items asterisked, and the revised drawings. That is a template with four fields (revision letter, date, reviewer, list of changes), which is why the module can generate it.

Finally, the binder shows the failure modes to design against: a revision letter reused (O in 2012 and 2015), a run of revisions with no MDL copy on file (F through I), a package reissued under a suffixed letter (V, then V2), a drawing renamed in the records but not on the drawing (SK214-14A to SK214-16A, fixed in 2009), and the FAA's request that minor-change intervals not exceed six months, which nobody is tracking.

Data model

Approval lives on the pair: drawing revision + accepted MDL revision



Lines show which tables reference which. Shaded boxes are the heart of the model.

TSO filings data model · 9 tables

Nine tables, all prefixed cert_ in the schema to keep them together (the diagram drops the prefix for room). Every table carries the SkyNet standard columns: id uuid, created_at, created_by, updated_at, updated_by, is_active.

Table	Purpose	Key columns and rules
cert_certifications	One row per FAA approval Skybolt holds	cert_type (tsoa, stc, pma); number (TSO-C148, SA3285SO, PQ1554CE); standard_or_tc (TSO-C148; 3A12, 3A17); title; issuing_office; issued_on; amended_on[]; limitations text; status (active, superseded, surrendered); facility. Unique on (cert_type, number).

Table	Purpose	Key columns and rules
cert_controlled_drawings	The drawing as a durable identity across revisions	drawing_number (SK26P); title; product_family (cloc_2000, cloc_4000, zloc, aloc, platemount, grommet, retainer); mdl_page_number numeric (1, 1.2, 54.1); part_id uuid null → parts; notes. Unique on drawing_number.
cert_drawing_revisions	One row per revision-block line on a drawing	drawing_id; revision_letter (A, B, Orig); revised_on; description ("Add Plating Notation"); revised_by (NCB); change_class (minor, major, administrative); file document_id. Unique on (drawing_id, revision_letter).
cert_mdl_revisions	One Master Drawing List issue	certification_id; revision_letter (D … V); suffix (null, or 2 for V2); issued_on; nature (original, authorization, minor_change); status (draft, submitted, accepted, superseded, withdrawn); accepted_by_event_id → cert_faa_events; cofc_id; summary text. Unique on (certification_id, revision_letter, suffix). A trigger refuses a new letter that is not the next in sequence unless an override reason is recorded.

Table	Purpose	Key columns and rules
cert_mdl_revision_items	Which drawing revision is on which MDL	mdl_revision_id; drawing_revision_id; page_number as printed; is_changed (the asterisk); note ("Format", "Drawing Format. Mat Finish"). Unique on (mdl_revision_id, drawing_revision_id). Each drawing appears at most once per MDL.
cert_faa_events	Every letter out and every answer back	certification_id; mdl_revision_id null; direction (submission, response); event_type (application, revision_notice, supplement, acceptance, authorization_letter, query, meeting); event_on; faa_project_number (SP12253AT-Q); faa_office; faa_reviewer; our_signer; subject; outcome (pending, accepted, approved, rejected, superseded); due_on (reply expected by); notes.
cert_certificates_of_conformance	The CofC reissued with each MDL	mdl_revision_id; issued_on; signed_by; addressed_to (Atlanta ACO; East Certification Branch); document_id.

Table	Purpose	Key columns and rules
cert_test_reports	Qualification tests cited in a submission	lab (Constellation Technology, IMR Test Labs); report_number (A020458, A110021A); tested_on; test_type (tensile, shear, salt_spray); specimens text; result; mdl_revision_id null; drawing_id null; document_id.
cert_documents	Every file: letters, stamped MDLs, CofCs, test reports, drawing PDFs, emails	owner_table + owner_id (polymorphic, constrained by CHECK to the cert_ tables); document_type; title; s3_key; sha256; page_count; scanned_from ("Ned TSO Binder, Section 54"); uploaded_by/at.

Two views make the data useful without joins in the UI:

- cert_current_approvals: for each controlled drawing, the latest accepted MDL revision it appears on, the drawing revision at that MDL, and whether a newer drawing revision exists that has not yet been submitted. This is the "is this drawing's current revision FAA-approved?" answer, and it is what the parts screens will show.
- cert_open_items: MDL revisions in draft or submitted with days outstanding; drawing revisions newer than the last accepted MDL; FAA events with a due_on in the past and no outcome; the days since the last accepted minor change against the six-month interval the FAA asked for in 2011.

The quality manual revision in force (QMS-00 Rev 004, approved January 14, 2026) is a certification-level fact; it is recorded as a cert_faa_events row of type acceptance against the TSOA with the manual revision in the subject, and does not need its own table in v1.

Linking to drawings and parts

This is the part Matt flagged as the eventual goal, and it is where the module pays for itself. Three identifiers are in play and they are not the same thing.

Identifier	Example	Lives in	Meaning
Drawing number	SK26P; SK213-2B; SK-GS	cert_controlled_drawings	A controlled design document. One drawing often defines a family of sellable parts (SK26P defines every SK26S51-( ) stud length)
SkyNet part	parts.id, parts.part_number	SkyNet parts table (drives work orders, BOMs, kiosks)	A manufacturable or purchasable item with a BOM and routing
Fishbowl part number	SK2600-4S; SK245-5B	Fishbowl part table, mirrored into SkyNet by the bridge	The sellable or stocked SKU, what appears on a sales order and a price list

The relationships are one-to-many at each step: a drawing defines many SkyNet parts; a sellable Fishbowl assembly consumes several SkyNet component parts. So the link is made at the component level and rolled up, never pasted onto the assembly by hand.

How the link is stored

- cert_controlled_drawings.part_id is nullable and points at the SkyNet part when the drawing defines exactly one part (SK-GS grommet, SK213 LOCKING CLIP). When a drawing defines a family, a bridge table cert_drawing_parts (drawing_id, part_id, length_or_dash) carries the many-to-many. The SkyNet BOM already knows which components make up a sellable assembly (explode_bom exists), so "is SK2600-4S a TSO article?" becomes: explode the BOM, look up each leaf's drawing, confirm every drawing's current revision is on the latest accepted MDL. That query is the heart of the sellable-part list Nadine asked for and that Roger needs to approve, and it will finally exist as a computed fact rather than a memory.
- Matching drawings to SkyNet parts is a one-time human job with tool help: the module proposes matches by prefix (SK26P → parts whose number starts SK26S51; SK213-2B → SK213-), and Compliance confirms or corrects each. Expect about 87 drawings to

map to a few hundred parts. Unmatched drawings (tooling, inserts that are never sold alone) stay unlinked and are reported as such.

What the parts screens gain

- A certification badge on the part record: TSO-C148 (MDL Rev V, SK26P Rev H), or Not a TSO article, or Pending (drawing Rev J submitted August 13, 2026, awaiting FAA).
- The same badge rolled up on the sellable assembly, with a warning when any component is pending or unlinked.
- A reverse lookup: from a drawing, every part and assembly it governs, which is what Ned needs when he changes a drawing and wants to know what it touches.

What this does not do

It does not make the module the source of truth for part numbers (Fishbowl remains that) or for BOM structure (SkyNet's BOM tables remain that). It adds one fact to each: the certification standing of the design behind it.

Document storage

The files are the evidence; the rows are the index. SkyNet already stores material certificates in S3 under material-certs/{receiving_id}/ with signed URLs minted in an edge function, and the same pattern applies here.

Rule	Detail
Bucket layout	certifications/{certification_number}/{document_type}/{yyyy}/{document_id}.{ext} — for example certifications/TSO-C148/mdl/2026/… and certifications/TSO-C148/drawing/SK26P/…
Immutability	A document row is never edited after upload; a corrected scan is a new row, the old one marked superseded_by. The sha256 is stored at upload so a file can be proven unchanged later
Versioning of drawings	Each drawing revision has its own PDF. The module never overwrites the Rev G PDF with Rev H; both exist, and the MDL item points at the one the FAA saw

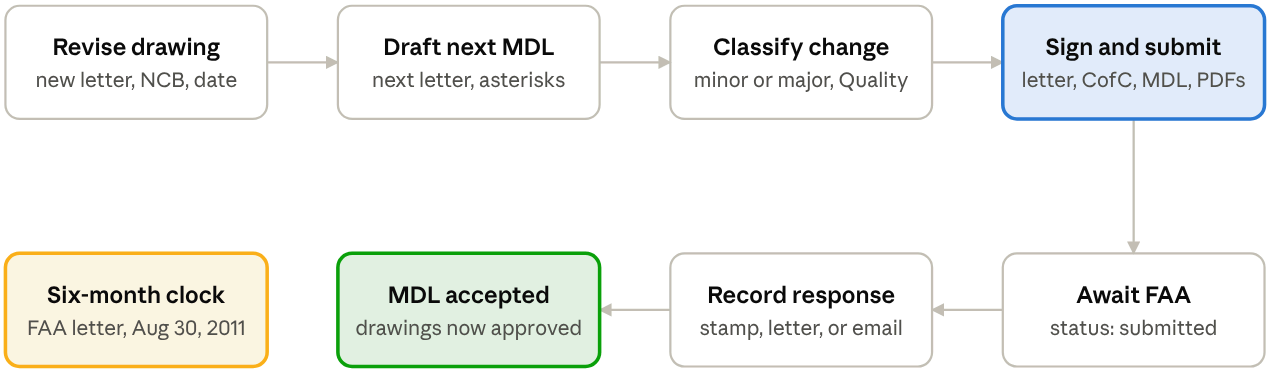
Rule	Detail
Access	Signed URLs only, generated server-side (the VITE_ lesson from earlier SkyNet work applies: nothing in the browser bundle). Read for Compliance, Admin, President, Viewer; upload for Compliance and Admin
Export control	Controlled drawings and test data are Skybolt technical data; they are ITAR/EAR-sensitive by nature even if not marked. The module labels every document with a sensitivity (public, customer_package, internal_technical). Only public and customer_package documents can be included in a generated customer approval package. Drawings are internal_technical by default and can never be exported by the package generator
Scans from the binder	Every page from Ned's binder is loaded as-is with scanned_from recorded, so the physical binder can be retired without losing provenance
Retention	Nothing is deleted. Superseded and withdrawn are statuses, not deletions

Size is not a concern: the entire binder is 56 MB, and the full TSO and STC archives together are under 1.5 GB even with their duplication.

Workflows

A drawing change is approved only after the FAA accepts the MDL it rides on

Minor-change workflow under 21.619(a), as the binder shows it from Revision O onward



The clock restarts at each accepted MDL; the module warns at five months if unsubmitted drawing revisions exist.

minor-change workflow · 7 steps and the six-month clock

1. Minor change (the common case, 21.619(a))

Ned revises a drawing and records the new revision in the module (letter, date, description, change class proposed as minor). The module shows the drawing as revised but unsubmitted. When Ned or Roger is ready, they open a new MDL revision; the module offers the next letter in sequence (refusing a reused one unless a reason is typed), pre-fills every current drawing revision, asterisks the changed ones, and lets them edit notes. Quality confirms the classification. The module generates the submission letter, the Certificate of Conformance dated the same day, the MDL as PDF, and a list of the changed drawing PDFs to attach; Ned signs the letter and CofC (wet or digital, his choice) and emails the package to the current FAA address on file. The MDL revision moves to submitted with a due_on of 30 days. When the stamped MDL comes back, Compliance records a response event, attaches the file, and marks the MDL accepted; the drawing revisions on it become approved and the parts badges update.

2. New design authorization (a new article)

As with SK40R17-2 in 2016 and the QL stud-nuts now in review: the module records an application event with the FAA project number, links the qualification test report, and holds the drawing revisions in a pending MDL that cannot be marked accepted until an authorization letter is attached. The FAA comment tracking sheets from the SK21037 review map directly onto a sequence of query and response events under one project number, so that history is preserved rather than living in Word files.

3. FAA response of any kind

Every inbound item becomes an event: acceptance stamp, authorization letter, query, change of reviewer, quality-system approval (QMS-00 Rev 004, February 2026). Events carry the reviewer and office so the contact history (Sprayberry, Bissell, Brito-Ross) is a query, not a memory.

4. The six-month clock

The FAA's 2011 letter asked that minor-change intervals not exceed six months. The module computes days since the last accepted minor change and, if any drawing revision exists that is newer than the last accepted MDL, shows a warning at five months and an overdue flag at six. If nothing has changed, no clock runs; the FAA did not ask for empty submissions.

5. Customer approval package

On request, the module assembles the current package from documents tagged customer_package: the latest accepted MDL with the FAA stamp, the authorization letters, the quality-system approval, the PMA letter, the current CofC, and the TSO standard. It writes a cover page with the date and MDL revision, and logs who requested it and who it went to. Drawings and test data are excluded by the sensitivity rule. This replaces the hand-assembled 2017 and 2021 packages and is what Sawyer sends a customer's quality department.

6. Annual review

Once a year, or before any FAA audit, Compliance runs the open-items view and the current-approvals view and signs a record that the file was reviewed. The binder, if Ned wants to keep it, can be reprinted from the module in MDL page order.

Roles, security, and audit

The module follows SkyNet's single-rule model (if you can see the tab, you can edit inside it) with two exceptions that the FAA context justifies: marking an MDL revision accepted, and generating a customer package, are gated to specific roles because each one is a representation to an outside party.

Action	Admin	Compliance	President (Ned)	Viewer / Customer Service	Everyone else
View certifications, drawings, MDL history, events	Yes	Yes	Yes	Yes	Part badge only
Add or edit a drawing revision	Yes	Yes	Yes	No	No
Draft an MDL revision; classify a change	Yes	Yes (classification is Quality's)	Yes	No	No
Generate the submission package	Yes	Yes	Yes	No	No
Mark submitted (records the outbound event)	Yes	Yes	Yes	No	No
Record an FAA response	Yes	Yes	Yes	No	No
Mark an MDL revision accepted (requires an attached FAA document)	Yes	Yes	No	No	No
Override the revision-letter sequence (with reason)	Yes	Yes	No	No	No
Generate a customer approval package	Yes	Yes	Yes	Request only	No
Upload documents	Yes	Yes	Yes	No	No
Change sensitivity of a document	Yes	Yes	No	No	No

Ned is the President role in SkyNet and today read-only in the main shell. This module is the one place where he does the work, so the President role gets write access to the cert_ tables specifically, enforced in RLS and not only in the UI. His signing authority is unchanged; the module records who signed, it does not sign for him.

Enforcement

- RLS on every cert_ table: SELECT for authenticated; INSERT and UPDATE for admin, compliance, president via user_has_role; DELETE for nobody (rows are deactivated, not deleted). This is stricter than the inventory tables, deliberately.
- The two gated actions (accept MDL, override letter) are SECURITY DEFINER RPCs that check role, require the FAA document id or the reason text, and write an audit row. The UI calls the RPC; it cannot flip the status column directly.

- audit_logs, SkyNet's existing table, receives a row for every status change, every generated document, and every customer package with who, when, and what. This is the trail an FAA inspector or an AS9100 auditor asks for.
- Documents are reached only through signed URLs minted by an edge function that checks role and sensitivity; drawing PDFs are never returned to a Viewer or Customer Service session.

Data handling rules carried into the module

- Controlled drawings and test data are Skybolt-owned technical data and stay inside SkyNet; the customer package generator cannot include them.
- No customer-owned data enters the module. Customer approval package requests record the customer's company name and the requester, nothing more.
- AI assistance (Claude in SkyNet, already integrated) may draft a submission letter or summarize a revision history, and the draft is marked as such; nothing AI-generated is sent to the FAA or a customer without a person's name on the event.

User interface

A new Certifications tab under Compliance in the SkyNet shell, visible to Admin, Compliance, President, and Viewer. Five screens, deliberately plain; this is a filing cabinet, not a dashboard.

Screen	What it shows	Main actions
Certifications	One card per approval: TSO-C148, each STC, each PMA; number, office, dates, status, current MDL revision, days since last accepted change, open items count	Open; add certification (Admin only)
MDL history	For one certification: every MDL revision as a row (letter, date, nature, status, submitted, accepted, reviewer), newest first; click a row to see its items with asterisks and the documents attached	New MDL revision; open a revision; download the MDL PDF as the FAA saw it

Screen	What it shows	Main actions
Drawings	The controlled drawing register in MDL page order: drawing number, title, family, current revision, approved revision, status badge (approved, pending, unsubmitted), linked parts count	Open a drawing; add revision; link parts
Drawing detail	Revision block as a table (letter, date, description, by, class, on which MDLs), the PDF for each revision, the parts and assemblies the drawing governs	Add revision; upload PDF; edit part links
FAA correspondence	Every event in date order with direction, type, project number, reviewer, subject, outcome, attachments; filter by certification, year, reviewer, open items	Record event; attach document; mark outcome

A sixth, Open items, is a view of cert_open_items pinned to the top of the tab: unsubmitted drawing revisions, MDL revisions awaiting the FAA with days outstanding, events past due_on, the six-month clock. Empty is the goal.

Generated outputs

Each is produced from the data and the templates on file; each carries a Generated by SkyNet footer with date, user, and MDL revision so a stray copy can always be traced.

Output	Template source	Format
Submission letter	The one-page letter in use since Revision O, with revision letter, date, reviewer, and address merged	Word and PDF (Word so Ned can edit before signing)
Certificate of Conformance	The current CofC wording, dated to the MDL, with signature block	Word and PDF

Output	Template source	Format
Master Drawing List	SKMDL TSO.xls layout: drawing, description, page, revision, date, notes, asterisks; header with revision, date, address	Excel (the FAA has always received it as a spreadsheet printout) and PDF
Submission package manifest	Checklist of letter, CofC, MDL, and each changed drawing PDF with file names and sha256	PDF
Customer approval package	Cover page plus the customer_package documents in a fixed order	Single PDF
Drawing register export	The Drawings screen as a spreadsheet, with approved revision and part links	Excel

On the part screens elsewhere in SkyNet

A single badge next to the part number, fed by cert_current_approvals: TSO-C148 · MDL Rev V (approved), TSO-C148 · pending Rev W (yellow), or no badge for parts outside any certification. Clicking it opens the drawing detail. On assemblies the badge reflects the weakest component.

Design notes

- Arial, Skybolt red for the single accent, grey rules, per the brand guide; SkyNet already follows this.
- Tables sort and filter in place; no modal wizards except the New MDL revision flow, which is three steps (pick letter and date, confirm items and asterisks, generate package).
- Every date is displayed with its source document one click away. In this module a date without a document is a claim, not a fact.

Initial data load

The binder and the two archives already contain everything the module needs for the TSOA. The load is mostly transcription with verification, in this order, so that each layer has what it references.

Step	What is loaded	From	Rows	Who verifies
1	The TSO-C148 certification row; the seven STC rows and two PMA rows as placeholders	FAA letters, certificates, PMA letters	10	Roger
2	The 87 controlled drawings with title, family, MDL page number	MDL Rev V (003 SKMDL TSO RevV 8-13-2026.xls)	87	Ned
3	Every drawing revision from the revision blocks on the drawings	Binder sections 1 through 56 (OCR-assisted, then eyeballed)	About 500	Ned, in MDL page order, a section at a time
4	MDL revisions D through V with dates and nature; items for each from the MDL copies on file	Binder MDL copies; archive xls files for Rev U and V	19 revisions, about 1,400 items	Compliance
5	FAA events: 19 submission letters, the FAA authorization letters and stamps, the 2015 and 2026 emails, the QMS approval	Binder Submission files, TSO Letters 2002-2016, archive	About 45	Compliance
6	Certificates of Conformance, one per MDL revision from O onward; test reports (A020458, A130554-A, A110021A, IMR salt spray)	Binder CofCs.pdf and Section 2; archive	About 15	Compliance

Step	What is loaded	From	Rows	Who verifies
7	Documents: every binder page and every authoritative archive file, attached to its row with scanned_from and sensitivity set	Binder scans; TSO and STC archives	About 150 files	Compliance
8	Drawing-to-part links	Proposed by prefix match against SkyNet parts; confirmed by hand	87 drawings to a few hundred parts	Ned and Roger
9	The two pending applications (SK21037/QL4S-ET, SK1820/QL8C20) as application events under SP19998AT-Q with their comment-sheet history	TSO archive	2 pending MDL drafts, about 20 events	Roger

Rules for the load:

- Dates and letters are entered exactly as the documents show them, anomalies included. Revision O appears twice; both are loaded, with the second flagged. F through I are created as placeholder rows with status unknown so the gap is visible rather than hidden.
- Where OCR read a scanned date, the loader records ocr_confidence and the row is not marked verified until a person has looked at the source page. Ned's binder sections are the only source for drawing revision blocks before 2013 and deserve his eyes.
- Nothing from the STC archive is loaded beyond the certification rows in Phase 1; the STC MDLs and kit records are Phase 3.
- Customer identities from the STC request folder are not loaded anywhere in this module.

The load scripts run against TEST first with the full dataset; Ned and Roger review on TEST; the same scripts run on PROD by a person after sign-off.

Phased delivery

Four phases, each usable on its own. No dates until the decisions below are made; the sizing is in working sessions, assuming Claude builds and Matt, Ned, and Roger review.

Phase	Scope	Deliverables	Done when	Size
1. Schema and load	The nine cert_ tables, two views, RLS, the two gated RPCs, audit hooks; load steps 1 through 7 for the TSOA; document storage	Migration file 2026-11-xx_cert_filings_phase1.sql applied to TEST; load scripts; a read-only Certifications tab showing the loaded data	Ned can find any MDL revision, drawing revision, or FAA letter from 2002 to today in SkyNet TEST faster than in the binder, and agrees the data is right	3 to 4 sessions, plus Ned's verification time
2. Working module	New MDL revision flow, drawing revision entry, FAA event entry, generated letter, CofC, MDL, and package manifest; open-items view; six-month clock	The tab becomes writable for Admin, Compliance, President; templates approved by Ned and Roger	The next real minor change (Rev W) is submitted to the FAA from the module, with the package generated by it and signed by Ned	3 sessions
3. Parts and STCs	Drawing-to-part links (step 8); part badges on SkyNet part and assembly screens; cert_current_approvals exposed to the sales dashboard; STC and PMA data loaded (STC MDLs, kits, Authorization for Use templates, airframe register)	Sellable TSO article list generated from the BOM, reviewed by Roger; STC tab with the seven certificates and their kits	Sawyer can answer "is this part a TSO article?" and "is this airframe on an STC?" from SkyNet without asking Ned	4 sessions

Phase	Scope	Deliverables	Done when	Size
4. Customer-facing and housekeeping	Customer approval package generator; annual review record; export of the binder in page order; Authorization for Use letter generator with per-aircraft dating and the airframe register; retire the paper binder as the file of record	Package and letter generators; review workflow; archive folder cleanup list acted on	A customer package and an STC authorization letter are each produced from SkyNet and sent, and Ned agrees the binder can become a backup	2 to 3 sessions

Each phase goes to TEST (ylzmyjjqibpbqbwjsnqj) first, where Claude can apply migrations and load data directly and will state what it ran and the result. Nothing touches PROD (luzungoqfuplspzbqctb) until a person applies the reviewed migration after it has passed on TEST, per the standing rule. Phase 1 is the only one with schema risk; phases 2 through 4 are UI, RPCs, and data.

What can start now, before any decision: the OCR pass over the binder's drawing revision blocks to produce a draft of step 3 for Ned to check. It is the longest pole and it depends only on the scans already in hand.

Decisions needed before Phase 1

1. **Ned's role.** Grant the President role write access to the cert_ tables (recommended), or have Ned work through Roger and Compliance with his own account read-only? This decides the RLS in the first migration. (Matt, Ned)
2. **Revision letter policy.** Keep the single-letter sequence and reserve a suffix for reissues (V, V2), as the binder already does, or move to letter plus date as the key? Recommended: letter plus optional numeric suffix, unique per certification, with the sequence trigger. (Ned, Roger)
3. **The two Revision O's.** Load both as O with the 2015 one flagged (recommended), or renumber the history? Renumbering changes what the FAA has on file and is not recommended. (Ned)
4. **Drawing files.** Does the module hold the released PDF only, or also the native CAD or Excel drawing file? Recommended: released PDF per revision in Phase 1; native files later if Ned wants them in one place. (Ned)

5. **Part linking owner.** Who confirms the drawing-to-part matches in step 8: Ned, Roger, or both? It touches what Sales may say about every part. (Matt)
6. **Template wording.** Ned and Roger approve the submission letter, CofC, and MDL templates as the current paperwork, or revise them while we are at it (the CofC still addresses "Atlanta ACO" on older copies; the 2026 one says East Certification Branch). (Ned, Roger)
7. **Six-month clock.** Treat the FAA's 2011 request as a standing rule the module enforces with warnings (recommended), or as history? (Roger)
8. **Customer package contents.** Confirm the document list and that drawings and test data are never included; confirm who may send a package. (Roger, Matt)
9. **Binder disposition.** After Phase 4, does the paper binder stay as Ned's working copy, become a printed backup refreshed annually from the module, or retire? Ned's call; the module supports all three. (Ned)
10. **Pending applications.** Load SK21037 and SK1820 into the module now as pending, so the FAA comment history is tracked there going forward, or wait until authorization? Recommended: now. (Roger)

Appendix: DDL sketch

A sketch of the core tables to settle names and keys before the migration is written. Column types follow SkyNet conventions (uuid primary keys, timestamptz, text with CHECK constraints rather than enums). Standard columns (created_at, created_by, updated_at, updated_by, is_active) and indexes are omitted for readability. The first migration file will be supabase/migrations/2026-11-xx_cert_filings_phase1.sql, applied to TEST by Claude with the result reported, and to PROD by a person.

```
-- Approvals Skybolt holds
create table cert_certifications (
  id uuid primary key default gen_random_uuid(),
  cert_type text not null check (cert_type in ('tsoa','stc','pma')),
  number text not null,                       -- 'TSO-C148', 'SA3285SO',
'PQ1554CE'
  standard_or_tc text,                        -- 'TSO-C148'; '3A12, 3A17'
  title text not null,
  issuing_office text,
  issued_on date,
  amended_on date[] default '{}',
  limitations text,
```

```
  status text not null default 'active' check (status in
('active','superseded','surrendered','pending')),
  facility text default '9000 Airport Blvd, Leesburg, FL 34788',
  unique (cert_type, number)
);

-- A drawing as a durable identity
create table cert_controlled_drawings (
  id uuid primary key default gen_random_uuid(),
  drawing_number text not null unique,       -- 'SK26P', 'SK-GS', 'SK213
LOCKING CLIP'
  title text not null,
  product_family text check (product_family in

('cloc_2000','cloc_4000','zloc','aloc','platemount','grommet','retainer','oth
er')),
  mdl_page_number numeric(6,2),              -- 1, 1.2, 54.1
  part_id uuid references parts(id),         -- when the drawing defines
exactly one part
  notes text
);

-- One revision-block line
create table cert_drawing_revisions (
  id uuid primary key default gen_random_uuid(),
  drawing_id uuid not null references cert_controlled_drawings(id),
  revision_letter text not null,             -- 'A' .. 'Z', 'Orig'
  revised_on date,
  description text,
  revised_by text,                           -- 'NCB'
  change_class text check (change_class in
('minor','major','administrative','unknown')),
  document_id uuid,                          -- the released PDF
  ocr_confidence numeric(3,2),
  verified_by uuid, verified_at timestamptz,
  unique (drawing_id, revision_letter)
);

-- One Master Drawing List issue
create table cert_mdl_revisions (
  id uuid primary key default gen_random_uuid(),
  certification_id uuid not null references cert_certifications(id),
  revision_letter text not null,             -- 'D' .. 'V'
```

```
  suffix smallint,                          -- null, or 2 for 'V2'
  issued_on date not null,
  nature text not null check (nature in
('original','authorization','minor_change')),
  status text not null default 'draft' check (status in
    ('draft','submitted','accepted','superseded','withdrawn','unknown')),
  accepted_by_event_id uuid,                -- fk to cert_faa_events, added
after that table
  summary text,
  sequence_override_reason text,
  unique (certification_id, revision_letter, suffix)
);

-- Which drawing revision sits on which MDL
create table cert_mdl_revision_items (
  mdl_revision_id uuid not null references cert_mdl_revisions(id),
  drawing_revision_id uuid not null references cert_drawing_revisions(id),
  page_number text,                         -- as printed
  is_changed boolean not null default false, -- the asterisk
  note text,
  primary key (mdl_revision_id, drawing_revision_id)
);

-- Every letter out and every answer back
create table cert_faa_events (
  id uuid primary key default gen_random_uuid(),
  certification_id uuid not null references cert_certifications(id),
  mdl_revision_id uuid references cert_mdl_revisions(id),
  direction text not null check (direction in ('submission','response')),
  event_type text not null check (event_type in

('application','revision_notice','supplement','acceptance','authorization_let
ter',
     'quality_system_approval','query','query_response','meeting','other')),
  event_on date not null,
  faa_project_number text,                  -- 'SP12253AT-Q'
  faa_office text,
  faa_reviewer text,
  our_signer text,
  subject text not null,
  outcome text check (outcome in
('pending','accepted','approved','rejected','superseded','n_a')),
  due_on date,
```

```
  notes text
);

alter table cert_mdl_revisions
  add constraint cert_mdl_revisions_accepted_by_fk
  foreign key (accepted_by_event_id) references cert_faa_events(id);

create table cert_certificates_of_conformance (
  id uuid primary key default gen_random_uuid(),
  mdl_revision_id uuid not null references cert_mdl_revisions(id),
  issued_on date not null,
  signed_by text not null,
  addressed_to text,
  document_id uuid
);

create table cert_test_reports (
  id uuid primary key default gen_random_uuid(),
  lab text not null,
  report_number text not null,
  tested_on date,
  test_type text check (test_type in
('tensile','shear','salt_spray','other')),
  specimens text,
  result text,
  mdl_revision_id uuid references cert_mdl_revisions(id),
  drawing_id uuid references cert_controlled_drawings(id),
  document_id uuid,
  unique (lab, report_number)
);

create table cert_documents (
  id uuid primary key default gen_random_uuid(),
  owner_table text not null check (owner_table in

('cert_certifications','cert_controlled_drawings','cert_drawing_revisions','c
ert_mdl_revisions',

'cert_faa_events','cert_certificates_of_conformance','cert_test_reports')),
  owner_id uuid not null,
  document_type text not null,
  title text not null,
  s3_key text not null unique,
```

```
  sha256 text not null,
  page_count int,
  sensitivity text not null default 'internal_technical' check (sensitivity
in
    ('public','customer_package','internal_technical')),
  scanned_from text,
  superseded_by uuid references cert_documents(id),
  uploaded_by uuid, uploaded_at timestamptz default now()
);

-- Bridge for drawings that define a family of parts
create table cert_drawing_parts (
  drawing_id uuid not null references cert_controlled_drawings(id),
  part_id uuid not null references parts(id),
  length_or_dash text,
  primary key (drawing_id, part_id)
);
```

Views and RPCs to follow in the same migration

- cert_current_approvals (security_invoker): per drawing, the latest accepted MDL revision, the drawing revision on it, the latest drawing revision of any status, and a derived status of approved, pending, or unsubmitted.
- cert_open_items (security_invoker): the four open-item classes and the six-month clock.
- accept_mdl_revision(p_mdl_revision_id, p_event_id) and override_mdl_letter(p_mdl_revision_id, p_reason), both SECURITY DEFINER, role-gated to admin and compliance through user_has_role, each writing to audit_logs.
- A BEFORE INSERT trigger on cert_mdl_revisions that compares the new letter to the certification's latest and raises unless it is the next letter, a suffix of the same letter, or sequence_override_reason is set.
- RLS: enable on every cert_ table; SELECT to authenticated; INSERT and UPDATE where user_has_role(auth.uid(), 'admin','compliance','president'); no DELETE policy.

The parts(id) reference assumes the SkyNet parts table name; it will be confirmed against the TEST schema before the migration is written.